

Program : Diploma in Computer Hardware Engineering	
Course Code : 5158	Course Title: Network Administration lab-II
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- This course is designed to provide hands-on experience for the students in System administration on Windows platform and Networks.
- Students will be able to develop skills in switch and router configuration also which will help them to work as network administrator. Also they can switch over to any other platforms in future.

Course Prerequisites:

Topic / Description	Course Code	Course Title	Semester
Knowledge in computer and system software		Introduction to IT Systems	1
Basic knowledge in operating system		System Administration Lab	3
Basics of networking		Network Administration Lab-1	4

Course Outcomes

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO 1	Demonstrate Windows Server installation and setting up of active directory	11	Applying
CO 2	Demonstrate FTP,NFS, DHCP & Firewall configuration	11	Applying
CO 3	Demonstrate basic IOS & Router Management commands.	9	Applying
CO 4	Implementing VLAN and inter-VLAN routing.	11	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate Windows Server installation and setting up of active directory		
M1.01	Demonstrate Windows Server installation, Creating workgroups & domains, setting up Active Directory.	6	Applying
M 1.02	Demonstrate the working of user access control, permission setting, profile setting, creating backups.	5	Applying
CO2	Demonstrate FTP,NFS, DHCP & Firewall configuration		
M 2.01	Demonstrate FTP Server configuration.	4	Applying
M 2.02	Illustrate the NFS.	1	Applying
M 2.03	Demonstrate Printer management.	2	Applying
M 2.04	Illustrate the setting up of DNS and DHCP.	2	Applying
M 2.05	Experiment Firewall configurations.	2	Applying
	Lab Exam – I	1½	
CO3	Demonstrate basic IOS & Router Management commands.		
M 3.01	Demonstrate Router - IOS Basic & Management Commands	3	Applying
M 3.02	Configure Static Routing on Routers	3	Applying
M 3.03	Configure RIP Routing on Routers	3	Applying
CO4	Implementing VLAN and inter-VLAN routing		
M 4.01	Configure a VLAN	4	Applying

M 4.02	Configure Inter-VLAN Routing using Router	4	Applying
M 4.03	Configure Inter-VLAN Routing using Layer 3 Switch	3	Applying
	Lab Exam – II	1½	

*** * Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Implement a network with a Windows Server and 2 or more clients. Create users with proper delegations. Install & configure a printer for shared use in the network.
2. Implement a VLAN in the campus.

Text / Reference:

T/R	Book Title / Author
T1	Mastering Windows Server 2019: The complete guide for IT professionals to install and manage Windows Server 2019 and deploy new capabilities, 2nd Edition.
R1	Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016

Online Resources:

Sl.No	Website Link
1	https://www.microsoft.com/en-in/cloud-platform/windows-server`
2	https://www.moderndeployment.com/windows-server-2019-active-directory-installation-beginners-guide/
3	https://hostadvice.com/how-to/how-to-install-and-configure-windows-server-2019-and-project-honolulu/